ACEFA
AUSTRALIA-AOTEAROA CONSORTIUM FOR EPIDEMIC FORECASTING & ANALYTICS

Lecture 1
Introduction to modelling

Lecturer: A/Prof Freya Shearer

Prepared for the ACEFA EpiAnalytics Course at Australian CDC, Canberra, November 2026 Slides prepared by Freya Shearer, Oliver Eales, James McCaw, Alexandra Hogan

Lecture scope



- What are models? What is a “good” model?
- Examples of modelling use in infectious disease research and policy
- Types of models
- The modelling process
- Model complexity
- Model non-identifiability
- Uncertainty and sensitivity analysis

What are mathematical models?

- Mathematical models are conceptual tools that explain how a real-world process or system behaves.
- The properties of a mathematical model must correspond in some way to the real-world process or system but are a simplification.
- Infectious disease models represent in a simplified manner the many processes involved in the spread and control of disease.

Mathematical models in infectious disease research and policy



A “good” model is suited to its purpose—that is, it should be as simple as possible and no simpler.

(Occam’s razor or principle of parsimony)

Keeling MJ & Rohani P (2008) Modelling infectious diseases in humans and animals. *Princeton University Press*.

Why do we use mathematical models in infectious disease research and policy?

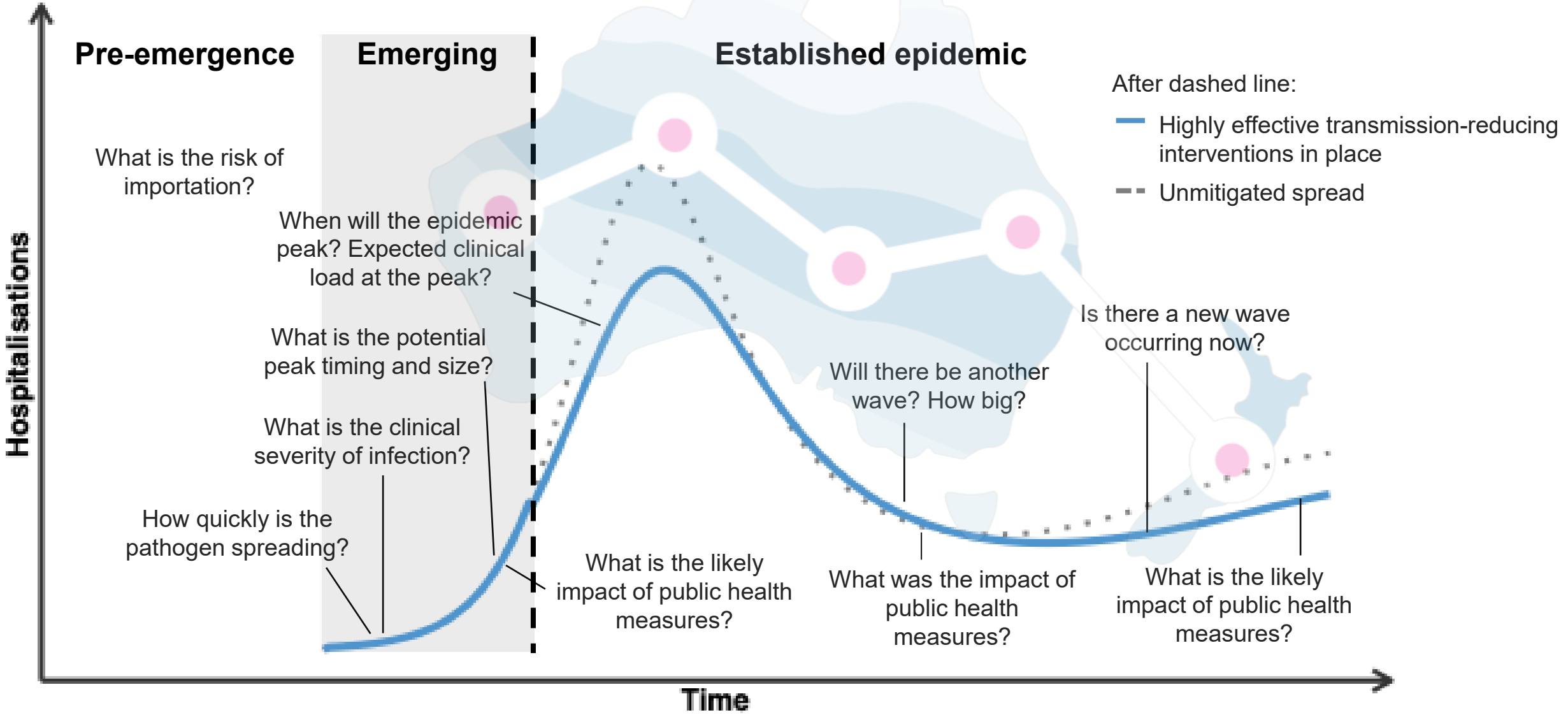


- Understanding and discovery:
  - Identifying key drivers of transmission
  - Estimating biological or epidemiological characteristics (e.g. R0)
  - Determining the relationship between multiple biological or intervention factors affecting transmission

- Analysis of trends in time:
  - Looking back: estimating past trends
  - Understanding the present: predicting current trends (“nowcasting”)
  - Looking forward:
    - Predicting short-term future trends (“forecasting”)
    - Exploring medium to long term “what-if” scenarios

- Other examples:
  - Analysis of trends in space (or time and space)
  - Importation/exportation risk assessment
  - Informing data collection (when/where/who/where given the objective and resources)
  - Identifying biases in surveillance indicators and study designs

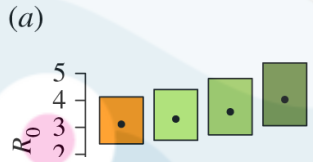
Example: pandemic response



Estimating pathogen characteristics

Novel coronavirus 2019-nCoV (COVID-19): early estimation of epidemiological parameters and epidemic size estimates

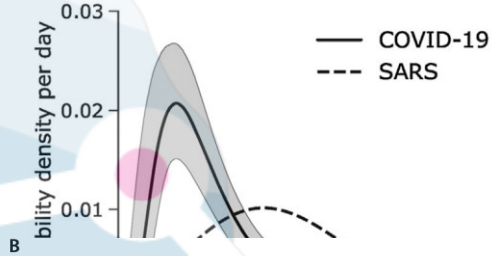
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Contents lists available at ScienceDirect

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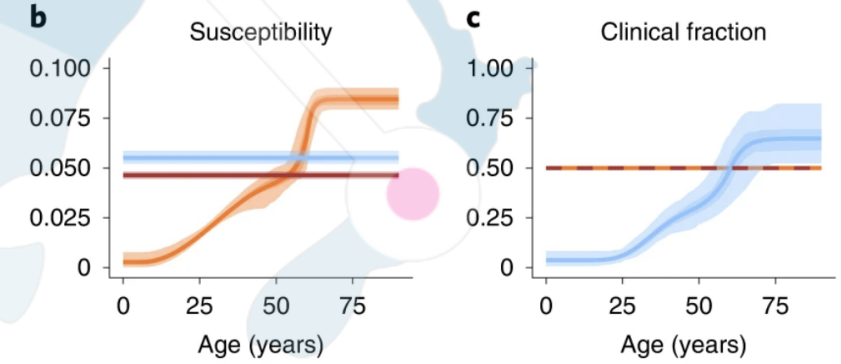
LETTERS

https://doi.org/10.1038/s41591-020-0962-9

Check for updates

Age-dependent effects in the transmission and control of COVID-19 epidemics

Nicholas G. Davies[1]✉, Petra Klepac[1,2], Yang Liu[1,2], Kiesha Prem[1], Mark Jit[1], CMMID COVID-19 working group and Rosalind M. Eggo[1]✉



The COVID-19 pandemic has shown a markedly low proportion of cases among children[1–4]. Age disparities in observed cases

cross-protection from other coronaviruses[9–11], or from non-specific protection resulting from recent infection by other respiratory

Why do we use mathematical models in infectious disease research and policy?

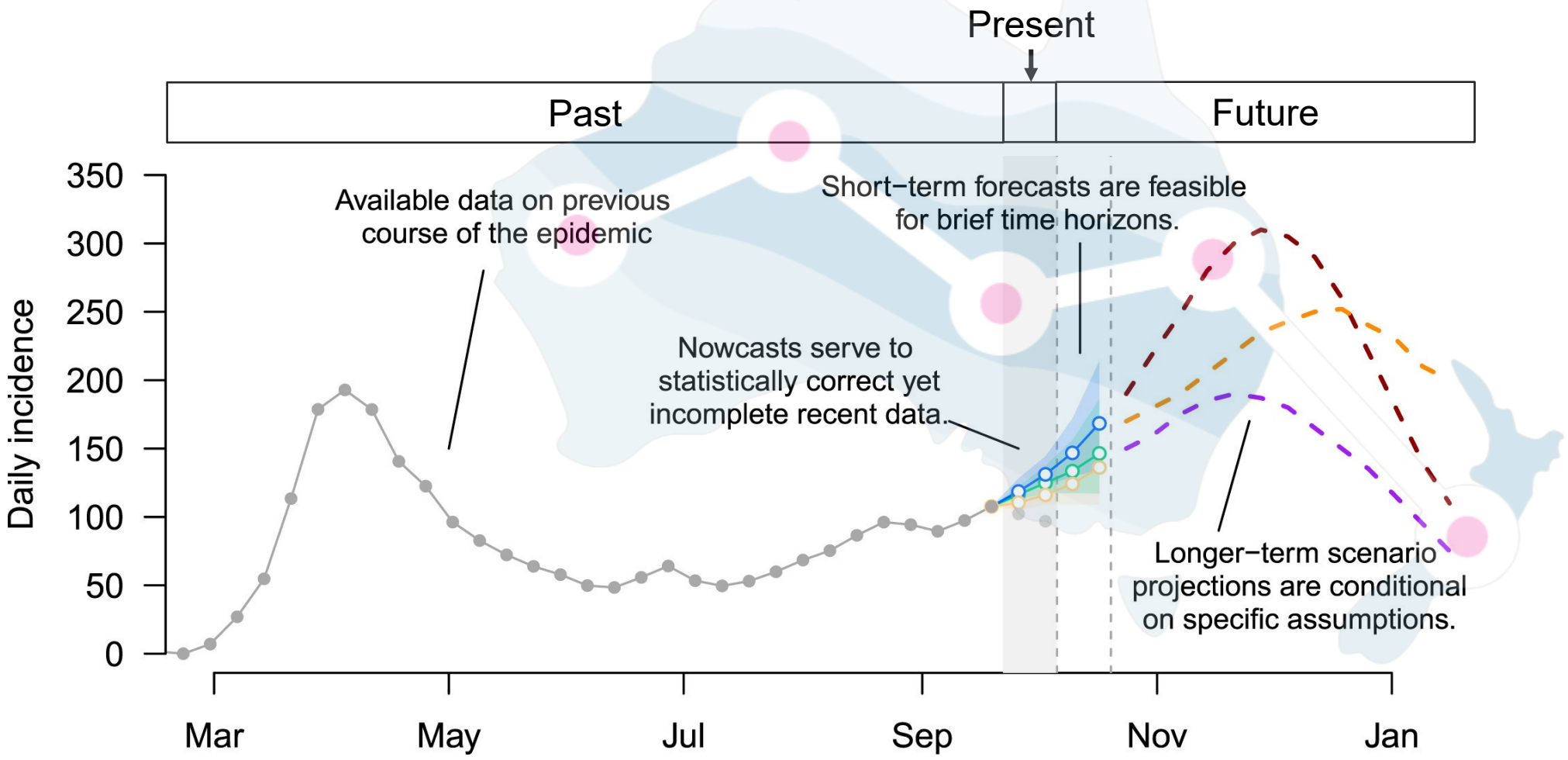


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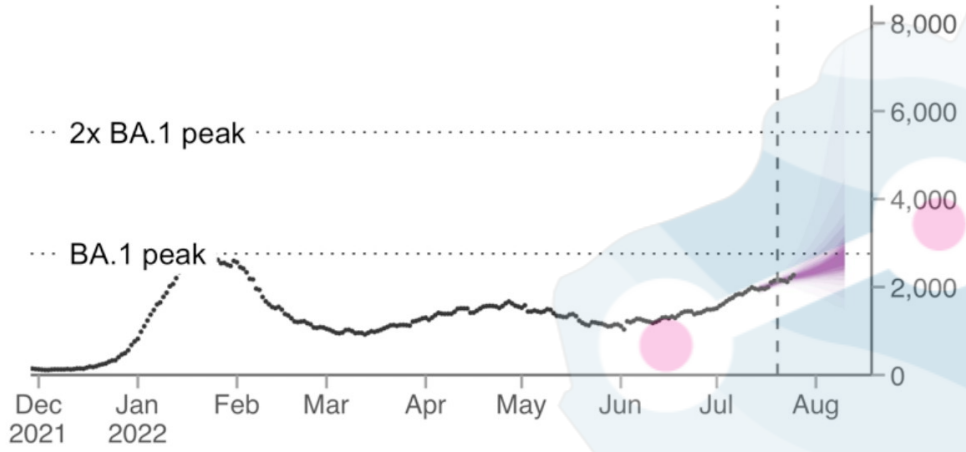
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Analysis of trends through time



Adapted from Wolffram et al 2026 *Int J Forecast*

Forecasting and Scenario Modelling for COVID-19

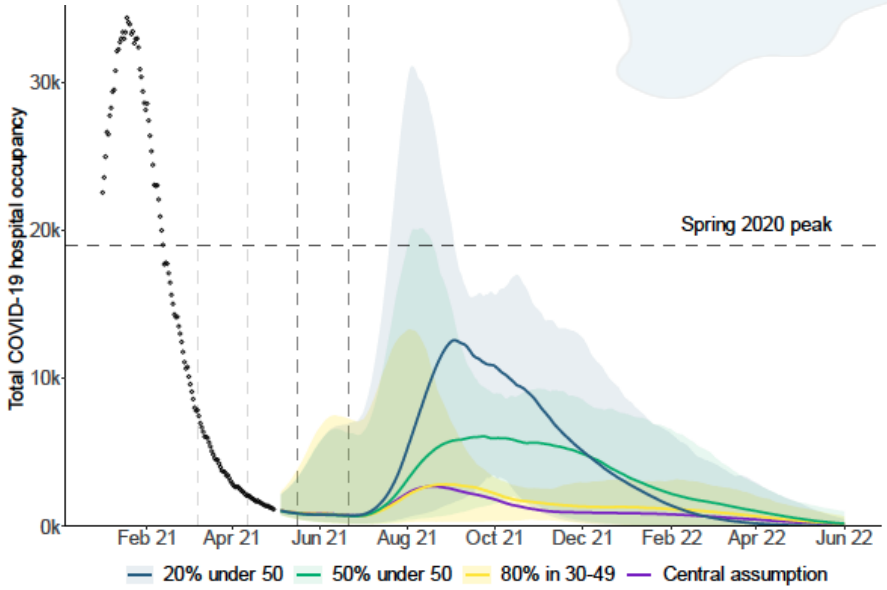


Hospital occupancy forecast for an Australian state.

Given current epidemic trends, if/when do we expect to exceed hospital capacity thresholds?

34% chance of exceeding by August 9th.

Analysis by Ruarai Tobin



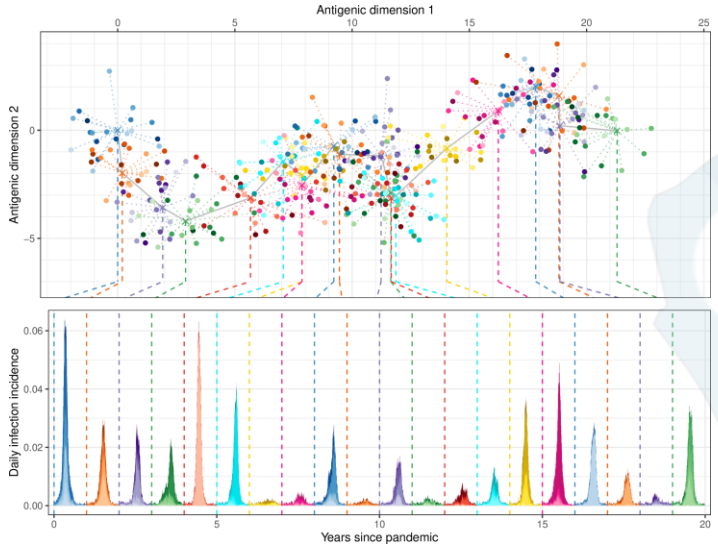
Vaccine rollout scenarios by SPI-M-O modelling teams for the UK Government.

(SPI-M-O = Scientific Pandemic Influenza Group on Modelling, Operational subgroup)

Analysis by CMMID COVID-19 working group

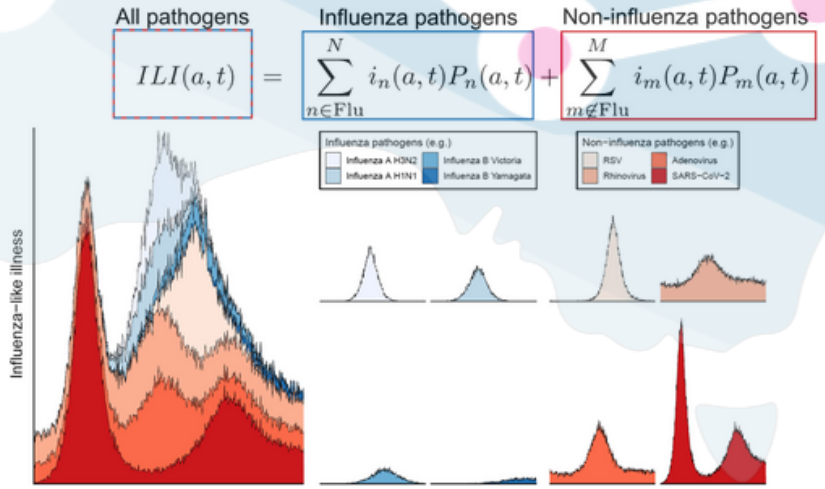
Many other types of questions and analyses

Exploring evolutionary immune dynamics of influenza



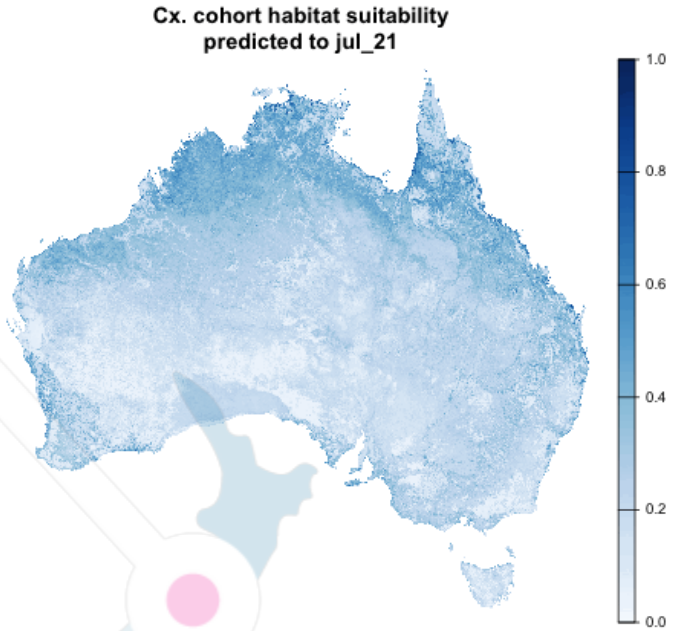
Eales et al (2025) *Plos Comp Biol*

Quantifying biases in surveillance indicators



Eales et al (2025) *Influenza Other Respir Viruses*

JEV vector habitat suitability in space and time



Duncan et al (2026) *Plos Negl Trop Dis*

Many types of models in infectious disease research and policy



- Classification by purpose:
  - Predictive versus explanatory
- Classification by mathematical characteristics:
  - Mechanistic (principle driven) or statistical (data driven)
  - Temporal or spatial (or both)
  - Deterministic or stochastic
  - Compartmental or agent-based (or both)

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Mechanistic versus statistical models



Mechanistic models (principal driven) describe how a process works, based on underlying physical or biological laws

E.g. SIR model describes how a susceptible person contacts an infectious person and becomes infected

Statistical models (data driven) describe what is observed in data without necessarily explaining the cause

E.g. GAM describes trends in a time-series of case notifications

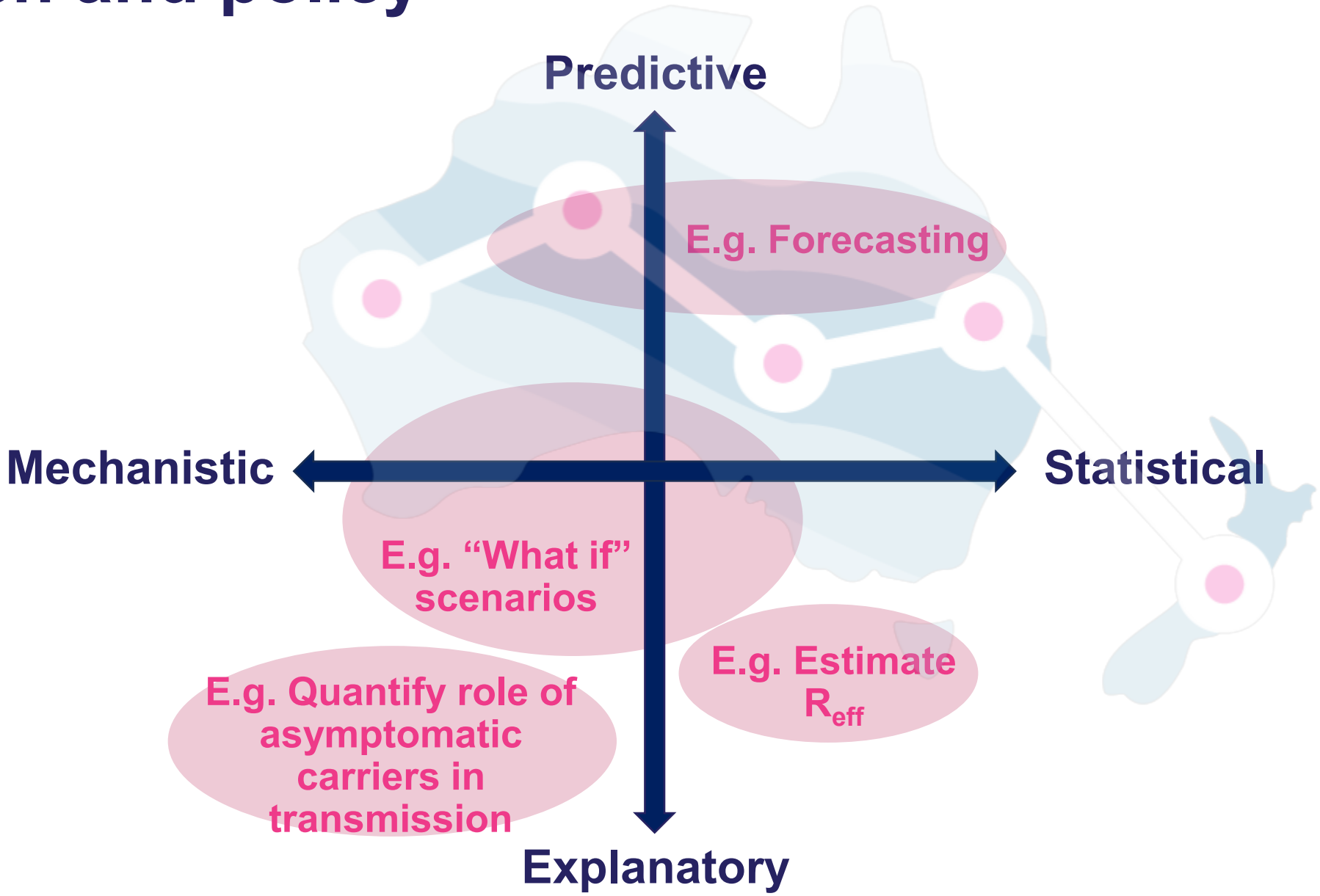
Models can be in-between!

Many types of models in infectious disease research and policy



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Many types of models in infectious disease research and policy



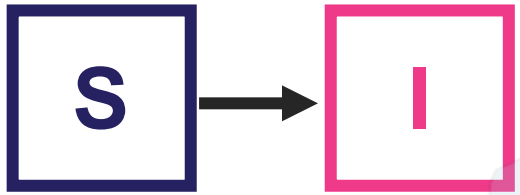
Many types of models in infectious disease research and policy



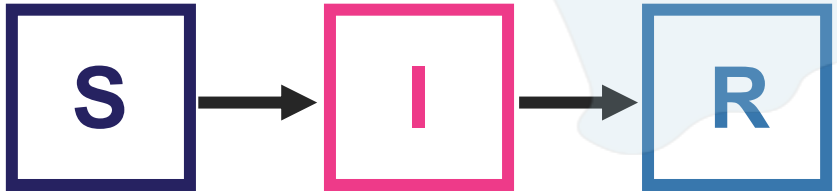
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- Classification by epidemiological characteristics:
  - Directly or indirectly transmitted (e.g. influenza vs dengue)
  - Recovery or lifelong infection (e.g. influenza vs HIV)
  - Partial or complete immunity (e.g. influenza vs measles)

Model classified by epidemiological characteristics

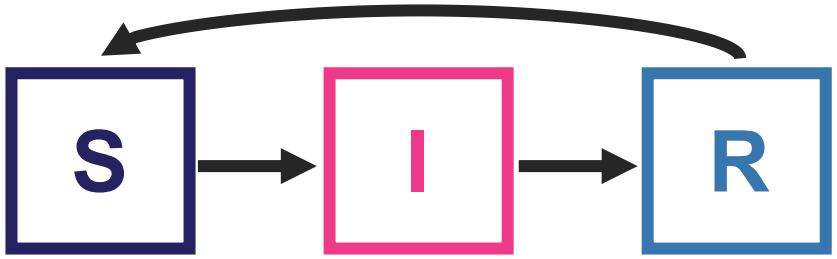
SI model (e.g. HIV)



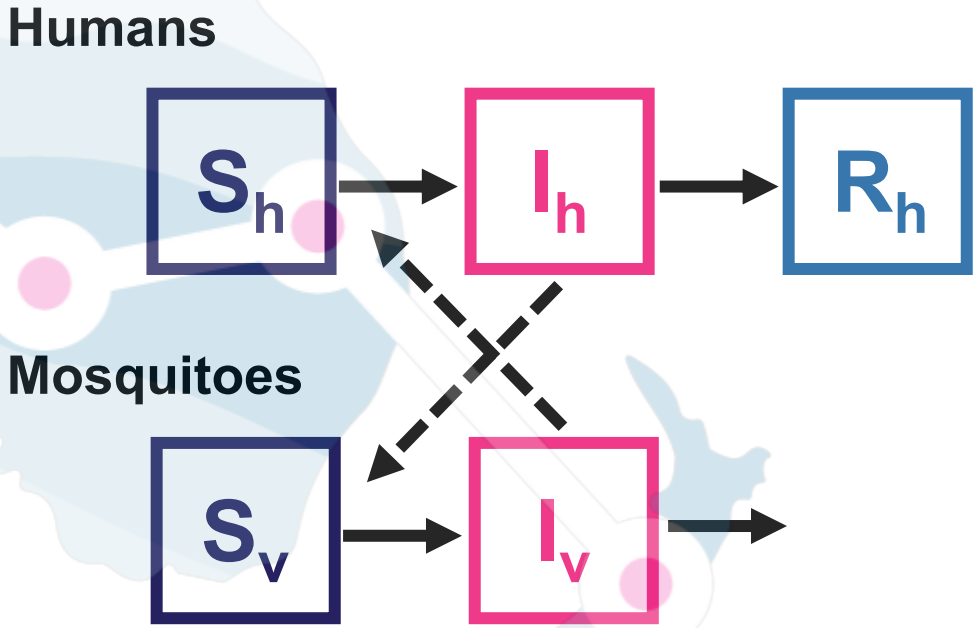
SIR model (e.g. measles)



SIRS model (e.g. influenza)



Ross-Macdonald model (e.g. dengue)



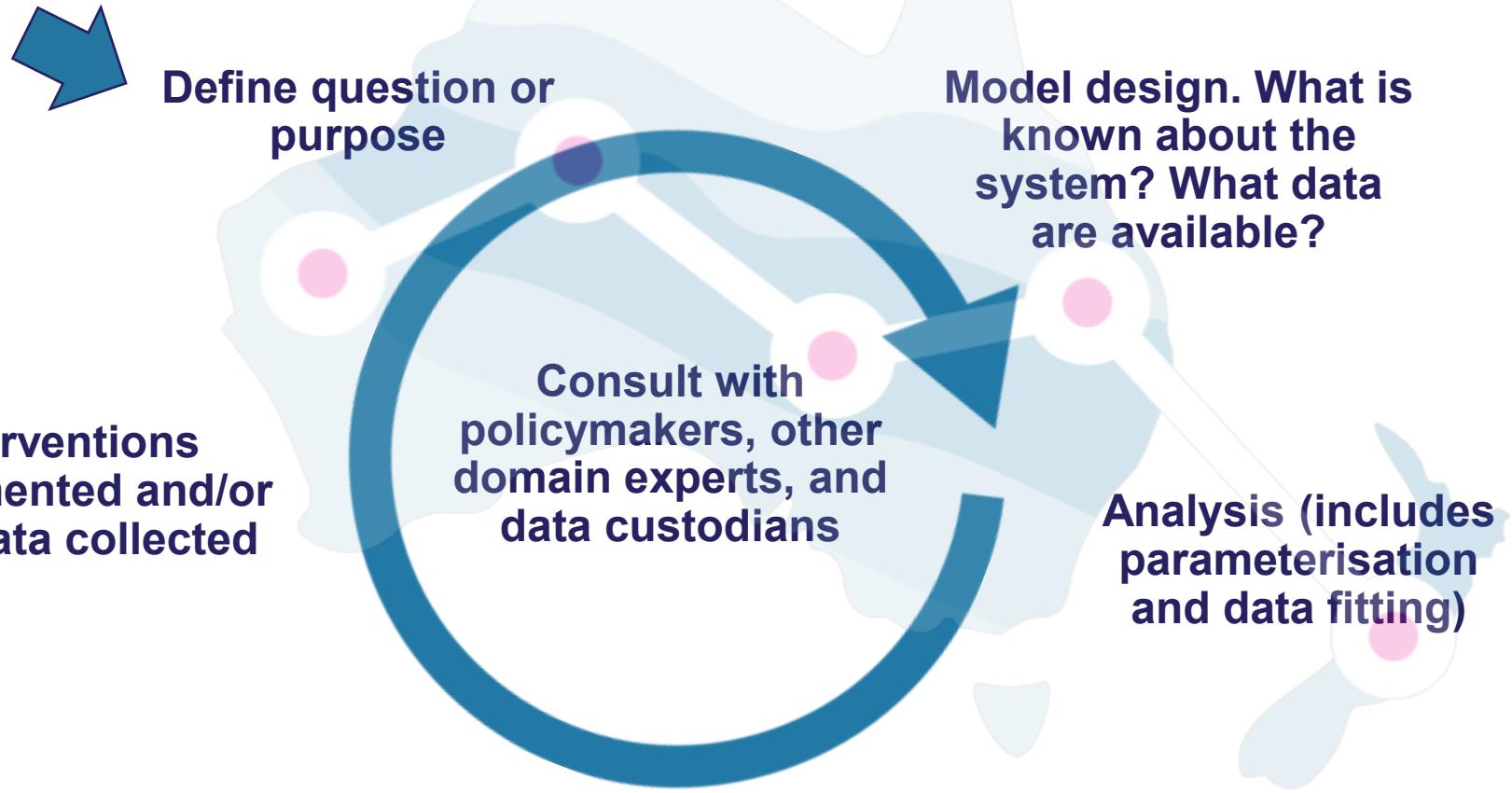
Many other biological and intervention features can be incorporated (see Lecture 3)

Mathematical models in infectious disease research and policy

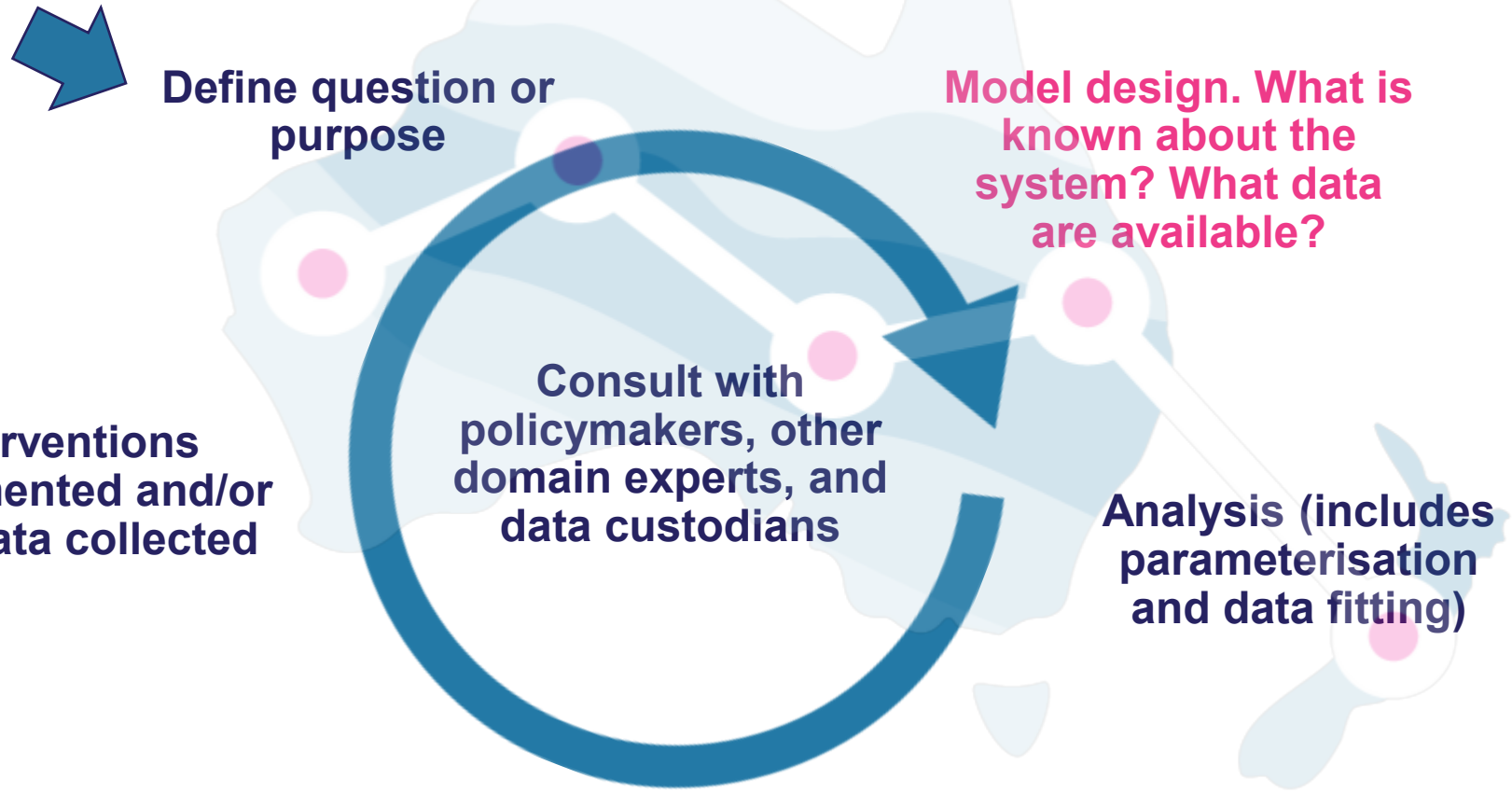


A “good” model is suited to its purpose—that is, it should be as simple as possible and no simpler.

An [idealised] modelling process



An [idealised] modelling process



Models must be fit-for-purpose, including an appropriate level of complexity



- Models range from simple (e.g. grouping individuals into compartments) or highly complex (e.g. simulating the behaviour of all individuals)
- The complexity of a model depends on:
  - The specific research question/purpose
  - Existing knowledge of the system
  - Available data
  - Timeframe for analysis and computational resources
- Model complexity is not generally an indicator of its utility
- Often a trade-off between reality (complexity, specificity) and understanding (simplicity, generalisability)
- More complex models require more detailed data for parameterization

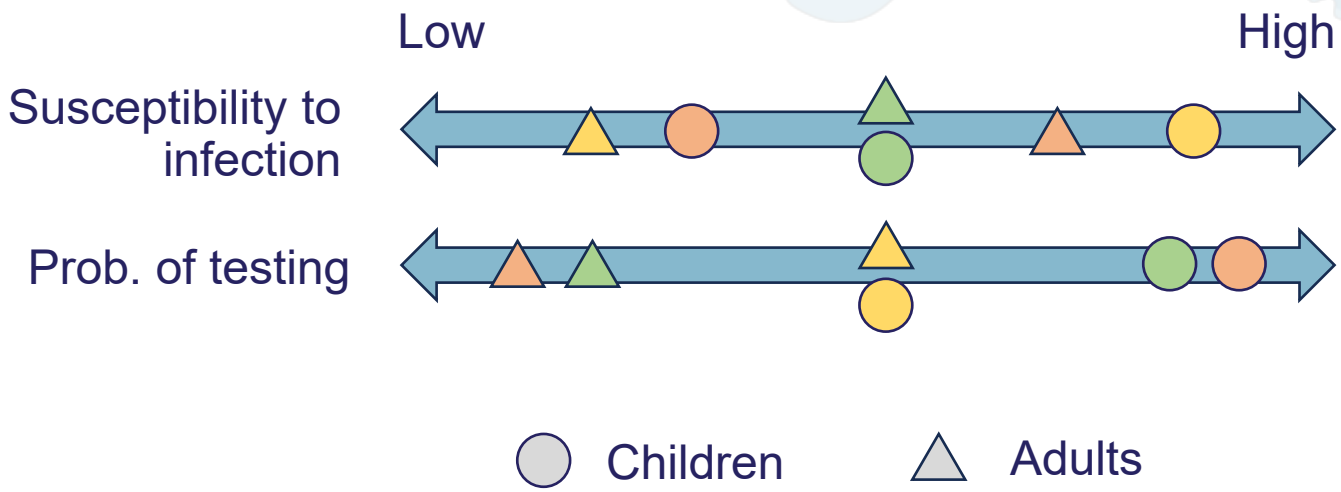
Model “non-identifiability”

When multiple different combination of model input parameters produce the same or very similar outputs.

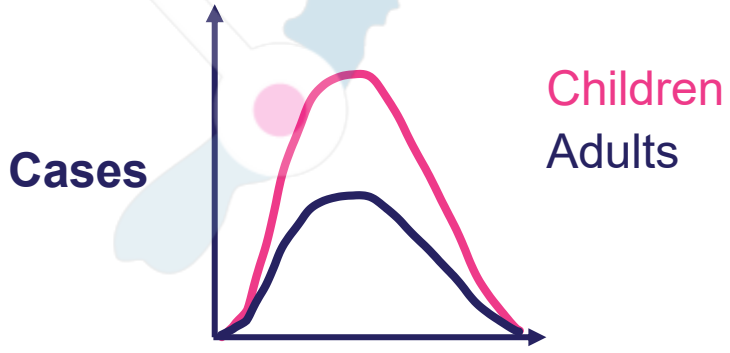
Policy question: What is the relative susceptibility to infection of children versus adults?

<u>Only</u> case data are available.

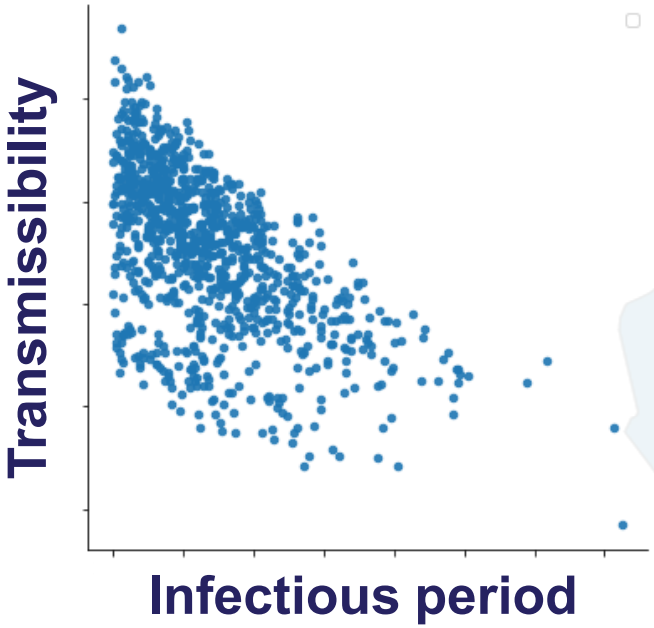
Model input parameters



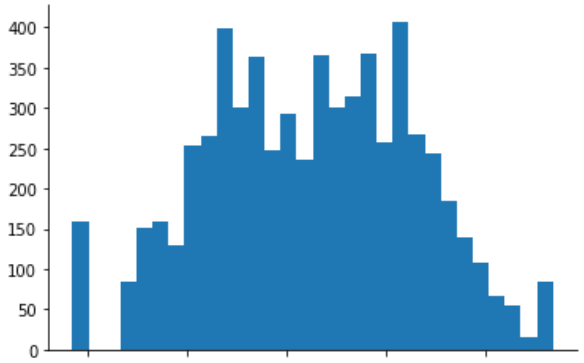
Case data (per 100,000)

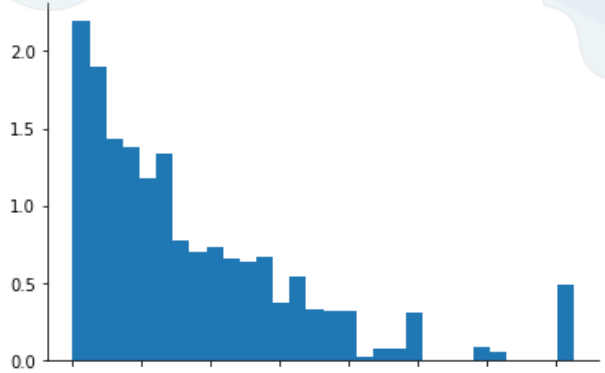


Model “non-identifiability” in practice

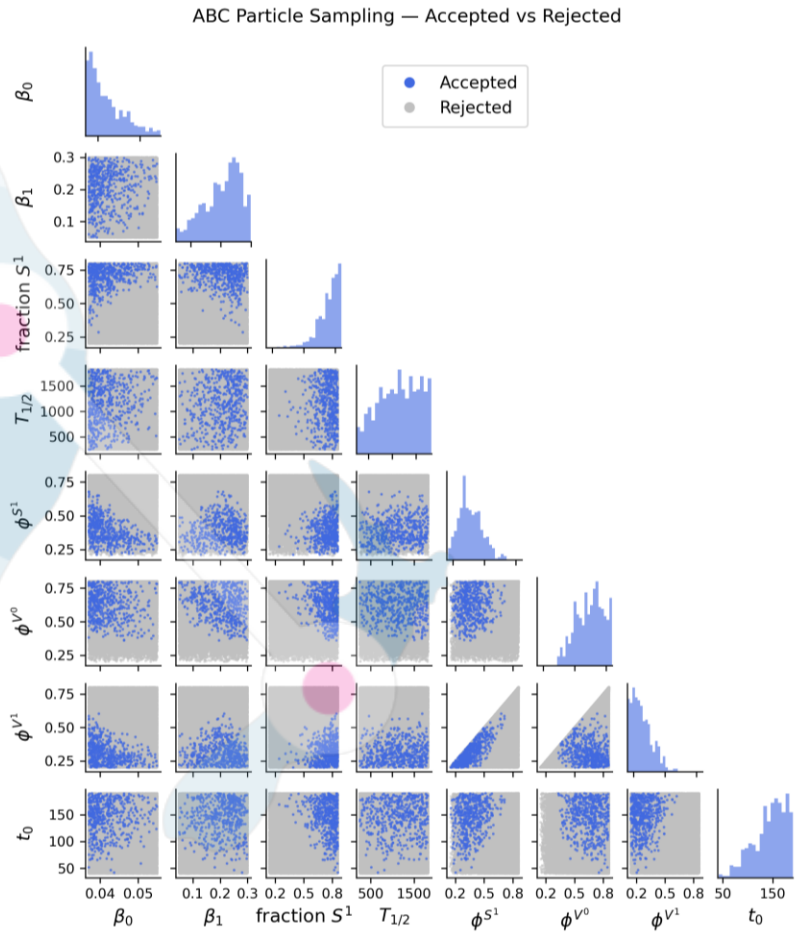


Each dot – model parameter combination selected via statistical inference/fitting procedure



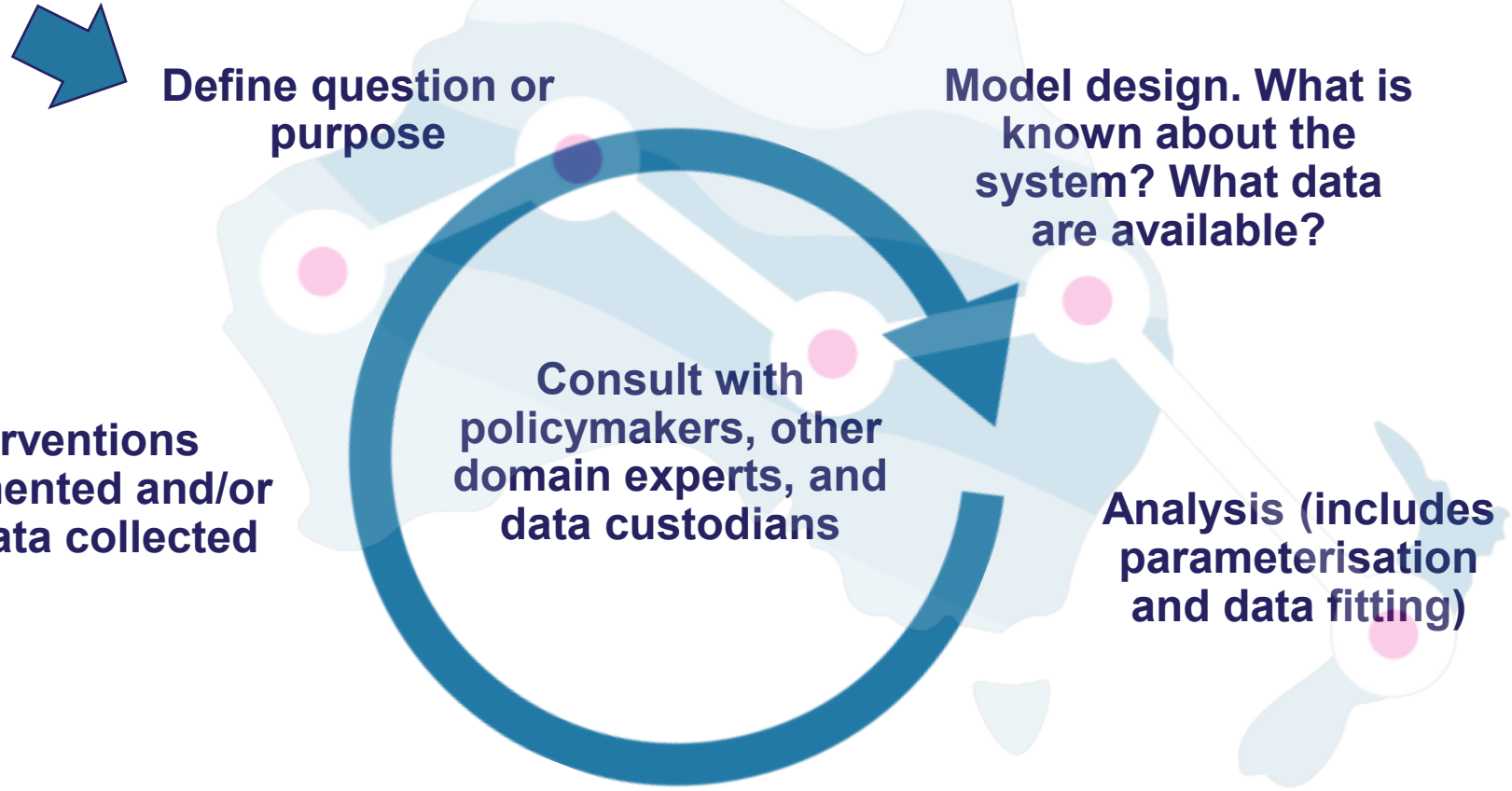


Analysis by Dr Marya Poterek (mpox spread in Australia)



Analysis by Dr Eliran Boksenbojm (influenza spread in Australia)

An [idealised] modelling process



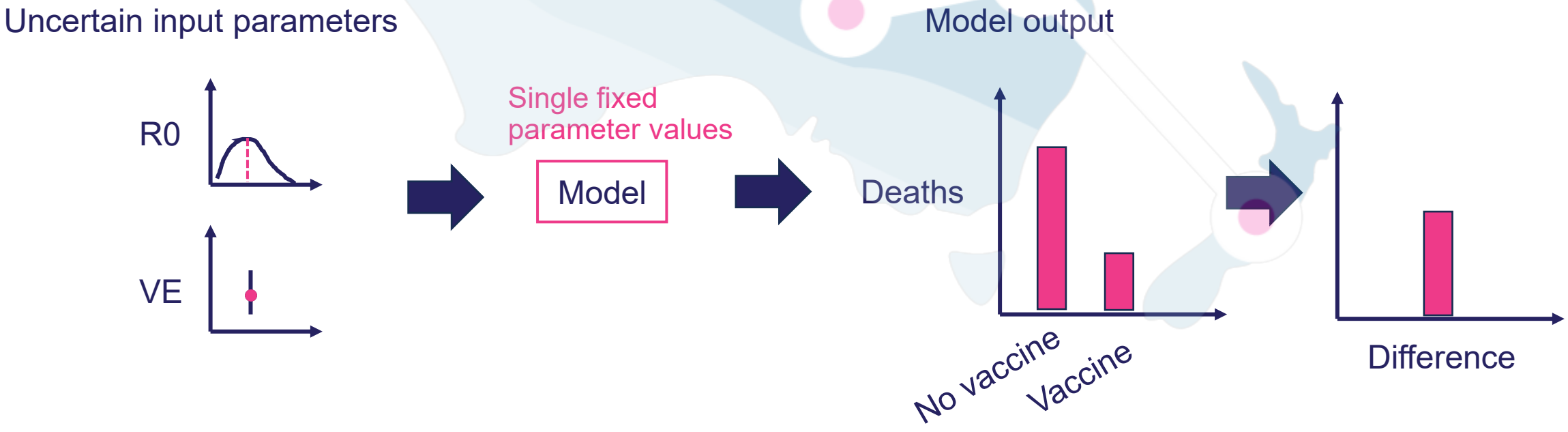
Uncertainty and sensitivity analysis

- **Uncertainty analysis — quantify the *range of model outputs* produced due to uncertainties in model inputs.**

- **Sensitivity analysis — determine which of the uncertain model inputs have the *greatest influence on model outputs (*and subsequent conclusions).**

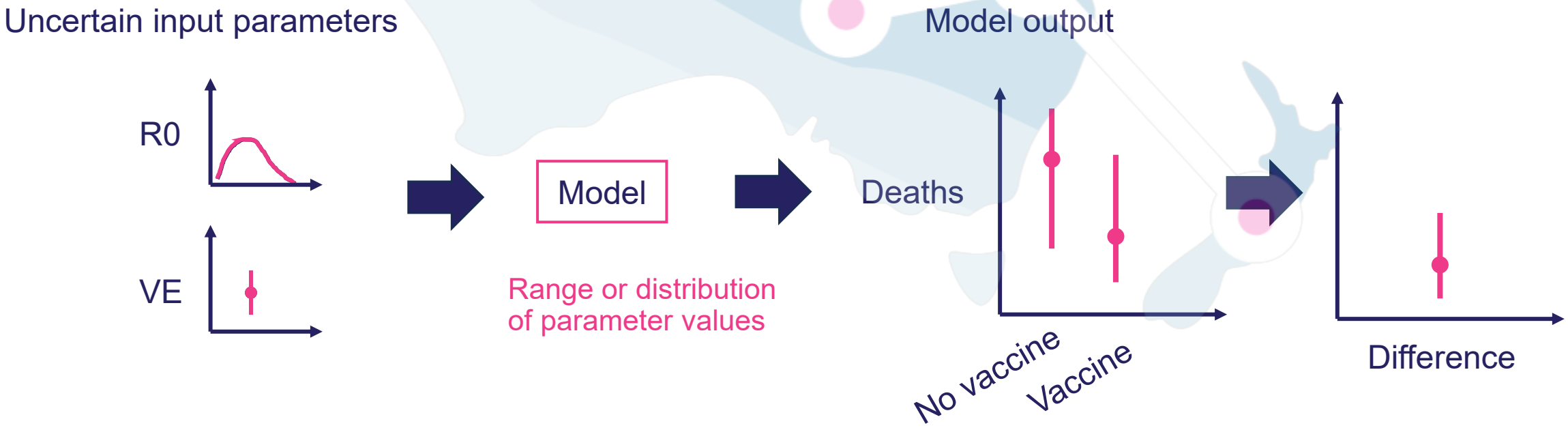
Uncertainty analysis

- **Quantify the *range of model outputs* produced due to uncertainties in model inputs.**
  - **Stochastic uncertainty**
  - **Parameter uncertainty**



Uncertainty analysis

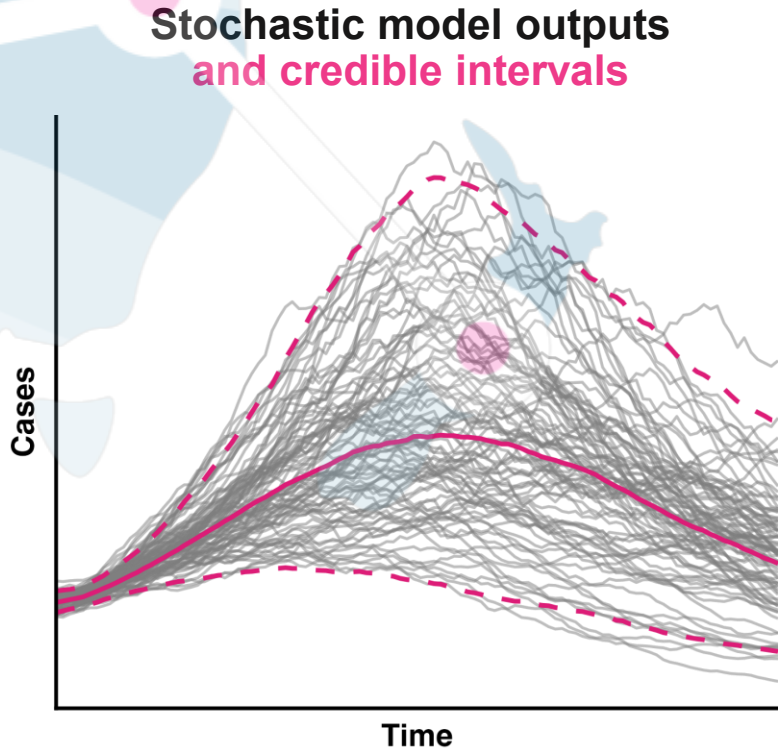
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Uncertainty analysis

- Quantify the *range of model outputs* produced due to uncertainties in model inputs.
  - Stochastic uncertainty
  - Parameter uncertainty

Side note: For stochastic models (covered later), the same input values will produce a range of model outputs, and so many simulations are required.

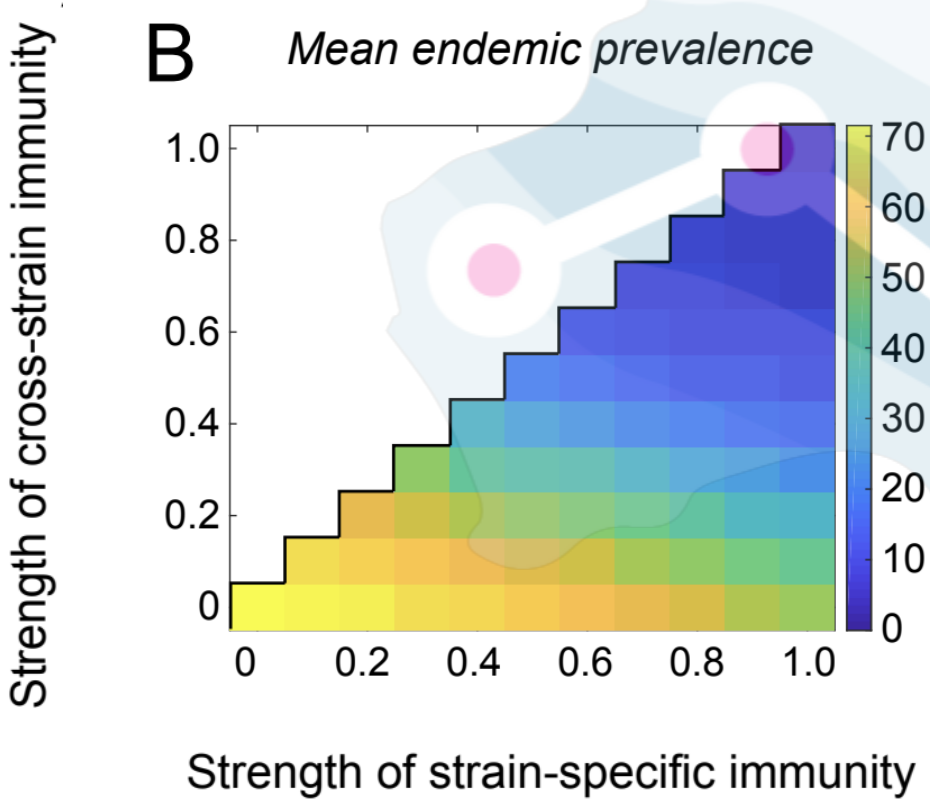


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Sensitivity analysis in practice



Two-way sensitivity analysis: influence of cross-strain and strain-specific immunity on prevalence

Chisholm et al 2021 (arXiv)

Summary



- Many types of models and modelling analysis
- A “good” model is suited to its purpose
- The first step of a modelling study is defining the purpose
- Model complexity is not generally an indicator of its utility
- Uncertainty and sensitivity analysis are important steps in a modelling analysis



Typical epidemic dynamics